# Emna Mastouri

emna.mastouri@isi.utm.tn | emnamastouri.github.io

## EDUCATION

**PhD in computer science** 2026 – present

*Faculty of Sciences of Tunis, LIMTIC laboratory*

- Configuring adaptive learning systems under uncertainty, with validity and stability constraints.

**Research Master's in computer science (software engineering)** 2024 – 2026

*Higher Institute of Computer Science (ISI), University of Tunis El Manar*

- Thesis: a neuro-symbolic approach to personalised mathematics tutoring for the Tunisian baccalaureate.

**Bachelor's degree in information and communication sciences and technologies** 2021 – 2024

*Faculty of Sciences of Tunis, University of Tunis El Manar*

**Baccalaureate in experimental sciences** 2021

*Lycée HHAW*

## EXPERIENCE

**Part-time lecturer** Ongoing

*Higher Institute of Computer Science (ISI), University of Tunis El Manar*

**End-of-studies internship** February – May 2024

*Tunisie Télécom*

- Developed a web application for monitoring computer networks.
- Technologies: Nagios XI, GNS3, VMware, Winbox, Angular, Node.js, MongoDB

**Internship** Summer 2022

*Tunisie Télécom, Central Service Management Unit, Kasbah*

- Maintenance of network equipment and transmission media.

## ACADEMIC PROJECTS

**Quality of LLM-generated vs. human-written code** *Python, SonarQube*

A comparison of code generated by language models with human-written code, looking at correctness, complexity and security warnings.

**Automated test generation and mutation testing** *Pynguin, mutmut*

Evaluation of two test generation tools on open-source projects, using code coverage and mutation score.

**Evolution of code smells in open-source projects** *PyDriller, SonarQube*

Tracking code smells across several releases of open-source projects, and studying how they relate to bug-fix commits.

## SKILLS

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| **AI and language models** | Python, LLM, RAG, multi-agent systems, SymPy, Streamlit |
| **Research tools (PhD)** | OR-Tools CP-SAT, MaxSAT, Knowledge tracing |
| **Software quality and testing** | SonarQube, PyDriller, Pynguin, mutmut |
| **Web development** | Angular, Node.js, MongoDB |
| **Networking** | Cisco (routers and switches), Nagios, GNS3, SD-WAN, IPv4 / IPv6, dynamic routing |
| **Embedded systems** | Arduino, ESP, STM, Raspberry Pi |
| **Personal strengths** | communication, interpersonal skills, adaptability, logical thinking, autonomy |

## LANGUAGES

Arabic: native | English: fluent | French: intermediate